

**SE 216 – SOFTWARE PROJECT MANAGEMENT**  
**SOFTWARE TOOLS DOCUMENT**

**PROJECT NAME:** AI Personal Health & Fitness Coach

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| TASK # | PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT                       |
|--------|-------------------------------------------------------------------------|
| 1      | Development of the AI Personalized Recommendation & Optimization Engine |
| 2      | Scalable Backend Infrastructure & Health Data Security Management       |
| 3      | Development Environment Setup for AI and Backend Implementation         |

**SOFTWARE TOOLS FOR TASK 1:**Development of the AI Personalized Recommendation & Optimization Engine

**Tool Cost/Training/Functionality Data**

| Tool          | Hugging Face      | vLLM              | AWS SageMaker | Azure AI Studio |
|---------------|-------------------|-------------------|---------------|-----------------|
| Cost          | \$0 (Open Source) | \$0 (Open Source) | \$500/mo      | \$420/mo        |
| Training Days | 5                 | 3                 | 12            | 10              |
| Functionality | 10                | 6                 | 8             | 9               |

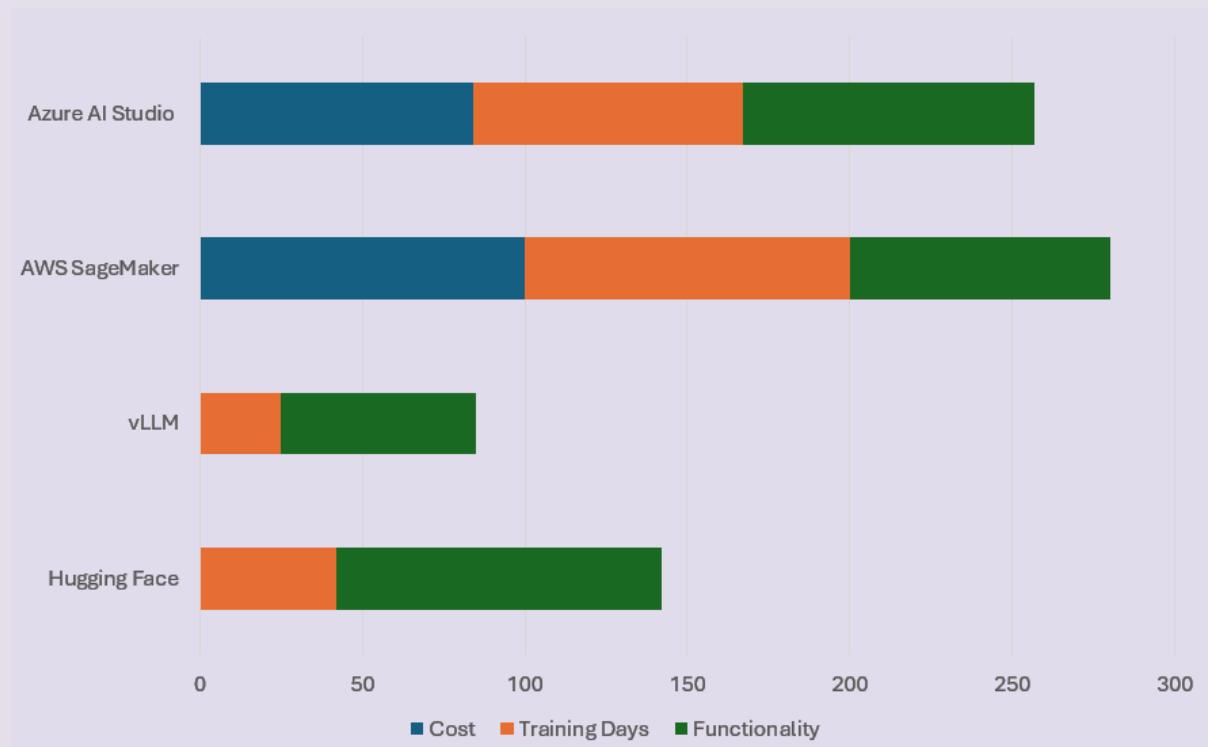
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#### Normalized Cost/Training/Functionality Data

| Tool          | Hugging Face | vLLM | AWS SageMaker | Azure AI Studio |
|---------------|--------------|------|---------------|-----------------|
| Cost          | 0            | 0    | 100           | 84              |
| Training Days | 42           | 25   | 100           | 83              |
| Functionality | 100          | 60   | 80            | 90              |

#### Normalized Tool Graph



Which tool has been selected? Why?

Hugging Face was selected.

Why: Getting pre-trained models up and running quickly was our top priority here, alongside painless fine-tuning. Hugging Face ended up being the obvious pick. Rather than spending almost two weeks wrestling with massive cloud environments like SageMaker just to deploy a tweaked model, their ecosystem lets us pull pre-trained weights, fine-tune them on our custom dataset, and get everything live in a fraction of the time. Plus, keeping the

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base cost at zero while still getting top-tier usability gives the team way more breathing room when we inevitably need to scale up the infrastructure later.

**SOFTWARE TOOLS FOR TASK 2:** Scalable Backend Infrastructure & Health  
Data Security Management

**Tool Cost/Training/Functionality Data**

| Tool          | Firebase | PostgreSQL      | MongoDB Atlas | Microsoft Azure SQL |
|---------------|----------|-----------------|---------------|---------------------|
| Cost          | 25/mo    | 0 (Open Source) | 30/mo         | 120/mo              |
| Training Days | 3        | 10              | 5             | 12                  |
| Functionality | 8        | 9               | 8             | 10                  |

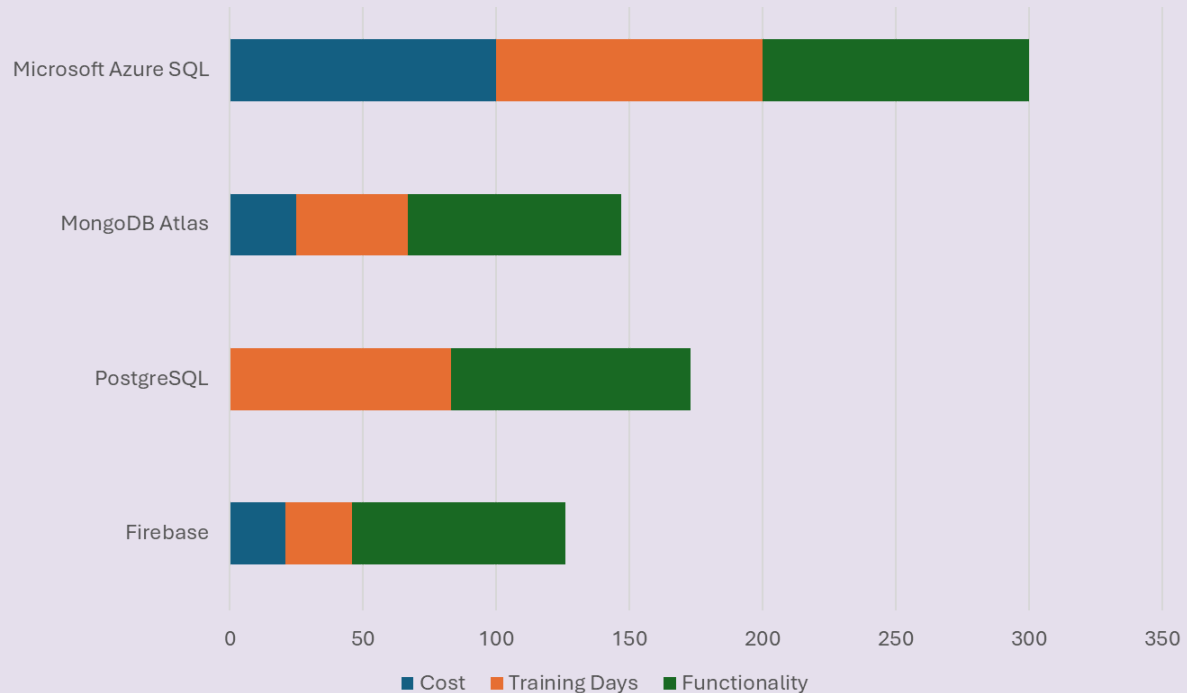
**Normalized Cost/Training/Functionality Data**

| Tool          | Firebase | PostgreSQL | MongoDB Atlas | Microsoft Azure SQL |
|---------------|----------|------------|---------------|---------------------|
| Cost          | 21       | 0          | 25            | 100                 |
| Training Days | 25       | 83         | 42            | 100                 |
| Functionality | 80       | 90         | 80            | 100                 |

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Normalized Tool Graph



**Which tool has been selected? Why?**

**PostgreSQL was selected.**

**Why:** PostgreSQL was selected for its strong ability to manage large-scale structured data, support complex queries, and ensure high data integrity. It guarantees that database operations are processed reliably and consistently, meaning that data is not lost, partially updated, or left in an inconsistent state even in case of system failures. This level of reliability is essential for handling sensitive health and fitness data. Additionally, PostgreSQL provides advanced features such as joins, indexing, and high scalability, making it a robust and efficient solution for backend development.

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**SOFTWARE TOOLS FOR TASK 3:**Development Environment Setup for AI and Backend Implementation

**Tool Cost/Training/Functionality Data**

| Tool          | PyCharm Professional | Wing IDE Pro | Eclipse | Visual Studio |
|---------------|----------------------|--------------|---------|---------------|
| Cost          | \$1500/yearly        | \$690        | \$0     | \$3240/yearly |
| Training Days | 6                    | 5            | 8       | 4             |
| Functionality | 10                   | 8            | 7       | 8             |

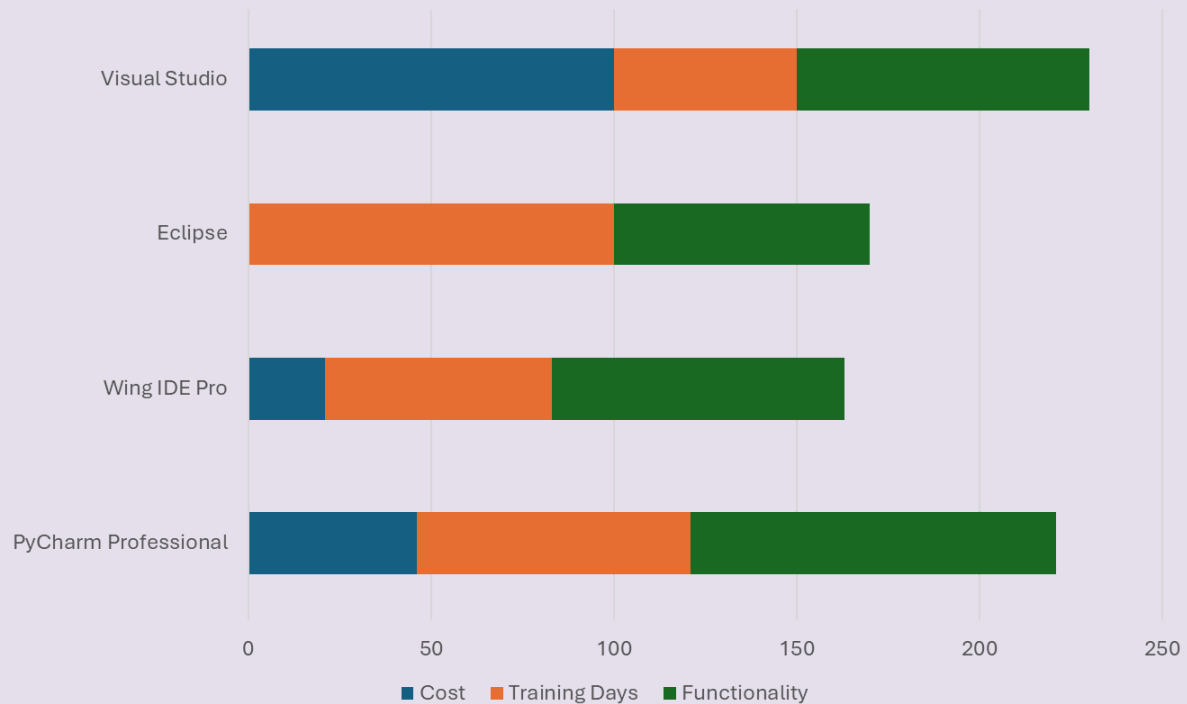
**Normalized Cost/Training/Functionality Data**

| Tool          | PyCharm Professional | Wing IDE Pro | Eclipse    | Visual Studio |
|---------------|----------------------|--------------|------------|---------------|
| Cost          | 46                   | 21           | 0          | <b>100</b>    |
| Training Days | 75                   | 62           | <b>100</b> | 50            |
| Functionality | <b>100</b>           | 80           | 70         | 80            |

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Normalized Tool Graph



#### Which tool has been selected? Why?

**PyCharm Professional** was selected as the development environment for Python because it provides advanced features such as intelligent code completion, powerful debugging tools, integrated testing support, database management, and seamless integration with AI and backend frameworks. Although it is a paid solution, its high functionality and enterprise-level capabilities make it the most suitable IDE for large-scale and AI-based projects.